

1. Consider the GBN protocol with a sender window size of 4 and a sequence number range of 1,024. Suppose that at time t , the next in-order packet that the receiver is expecting has a sequence number of k . Assume that the medium does not reorder messages. Answer the following questions:

- What are the possible sets of sequence numbers inside the sender's window at time t ? Justify your answer.
- What are all possible values of the ACK field in all possible messages currently propagating back to the sender at time t ? Justify your answer.

2. Consider the SDN OpenFlow network shown in Figure 1. Suppose that the desired forwarding behavior for datagrams arriving at S1 is as follows:

- any datagrams arriving on input port 4 from hosts h3 or h4 that are destined to hosts h5 or h6 should be forwarded over output port 1;
- any datagrams arriving on input port 4 from hosts h5 or h6 that are destined to hosts h3 or h4 should be forwarded over output port 4;
- any arriving datagrams on input ports 1 or 4 and destined to hosts h1 or h2 should be delivered to the host specified;
- hosts h1 and h2 should be able to send datagrams to each other.

Specify the flow table entries in S1 that implement this forwarding behavior

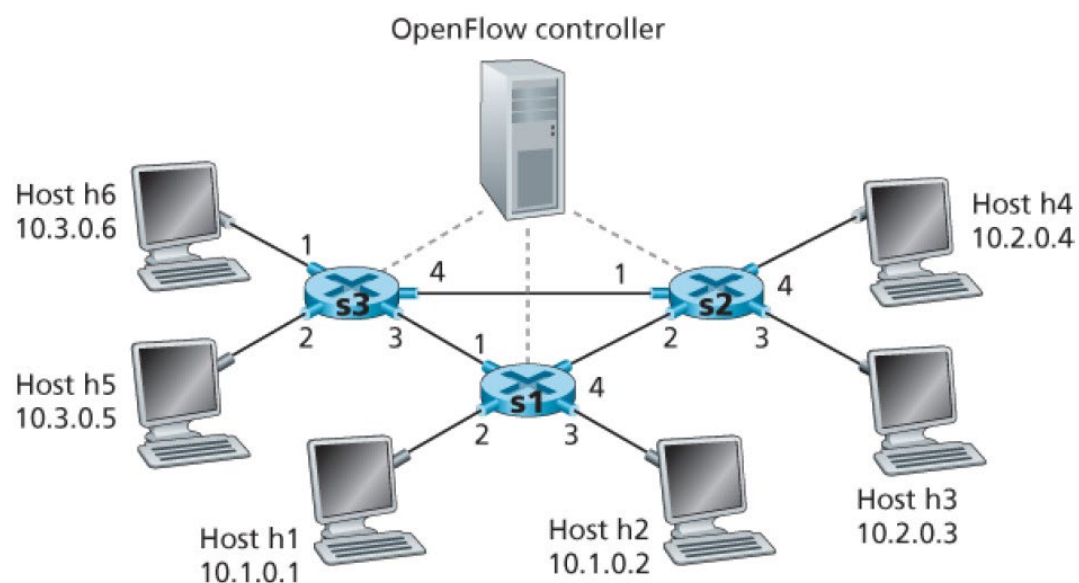


Figure 1

c. Give a flow table for S3 that implements the following firewall behavior:

Only traffic arriving from hosts h2 and h3 should be delivered to hosts h5 or h6. Arriving traffic from hosts h1 and h4 is blocked.

3. Consider the arrival of 10 packets to an output link at a router in some interval of time, as indicated by the table below. We'll consider time to be "slotted", with a slot beginning at $t = 0, 1, 2, 3$, etc. At the beginning of each time slot, the packet scheduler will choose one packet, among those queued (if any), for transmission according to the packet scheduling discipline (that you will

select below). Each packet requires exactly one slot time to transmit, and so a packet selected for transmission at time t , will complete its transmission at $t+1$, at which time another packet will be selected for transmission, among those queued. You can see the assignment of packets to classes 1,2, and 3 below.

Packet	1	2	3	4	5	6	7	8	9	10
time	0.35	0.85	1.25	1.68	1.97	2.2	2.93	3.24	3.55	3.95
class	2	2	2	3	2	1	2	3	3	2

- Schedule the transmissions using a priority policy, where class 1 has the highest priority and class 3 the lowest priority.
- Schedule the transmissions using a round-robin policy.